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1. TAKIM ŞEMASI

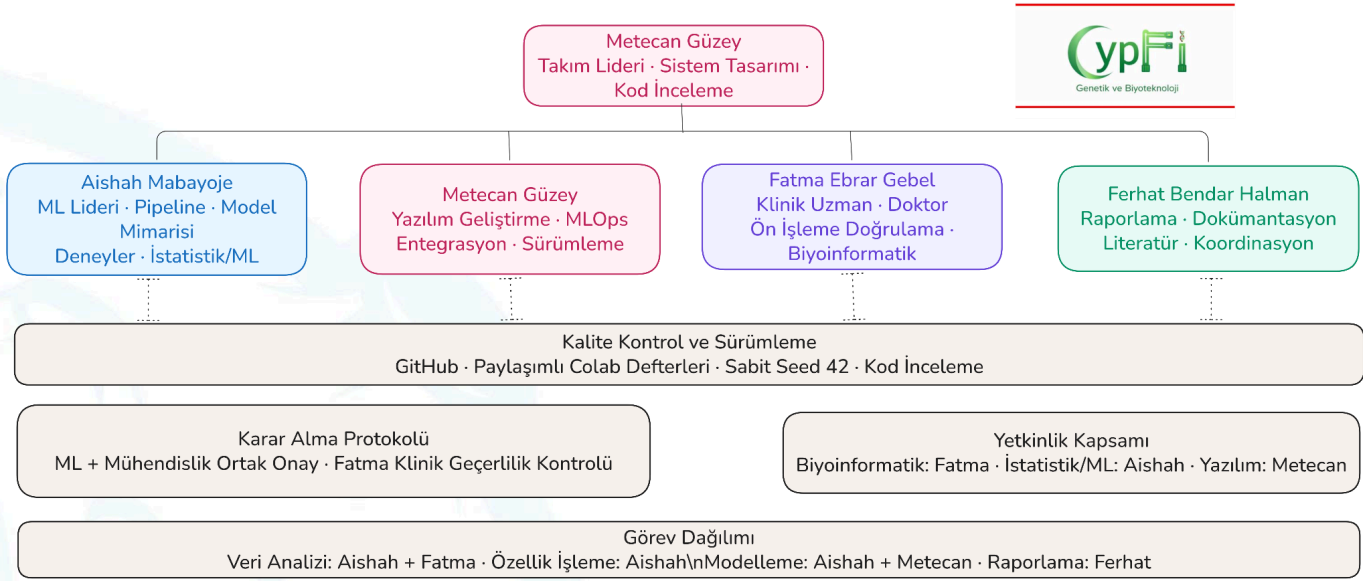


Fig 1: TAKIM ŞEMASI

2. PROBLEME EN YAKIN ÇÖZÜM SUNAN ULUSLARARASI MAKALELERİN ÖZETİ (10 PUAN)

Eight studies were selected to trace the methodological evolution of missense variant pathogenicity prediction. Each addresses (i) problem and data type, (ii) approach, (iii) data sources, (iv) reported metrics, and (v) key limitations.

[1] Ioannidis et al. — REVEL (2016, American Journal of Human Genetics) (i) Binary classification of rare missense variants as pathogenic or benign. (ii) Random Forest ensemble combining 13 in silico scores including SIFT and PolyPhen-2. (iii) HGMD and SwissVar for training; ClinVar for independent validation. (iv) AUC = 0.908, outperforming prior ensemble methods. (v) Single genome-wide threshold ignoring gene-specific context; requires all constituent score inputs to be available.

[2] Qi et al. — MVP (2021, Nature Communications) (i) Binary classification with explicit gene-level context. (ii) Deep residual network trained on 38 correlated features including conservation scores and population frequency, with separate models per LoF-tolerance class. (iii) HGMD, UniProt, DiscovEHR. (iv) AUC = 0.96 for constrained genes, 0.92 for non-constrained. (v) Requires complete feature matrix and large gene-specific training corpora — unavailable in blind-feature settings.

[3] Cheng et al. — AlphaMissense (2023, Science) (i) Genome-wide classification across all 71 million possible human missense substitutions. (ii) AlphaFold2 protein structure prediction combined with a protein language model fine-tuned on ClinVar. (iii) ClinVar, UniProt, primate sequence databases. (iv) AUC = 0.89 overall; precision

drops to 32.9% on rare disease cohorts. (v) Requires full protein sequences and AlphaFold2 structural coordinates — both unavailable in the competition format.

[4] McDonald et al. — AlphaMissense evaluated on CFTR (2024, PLOS ONE) (i) Clinical reliability of AlphaMissense specifically for CFTR missense variants. (ii) AlphaMissense scores benchmarked against established clinical severity measures from CFTR2.org. (iii) CFTR2.org, ClinVar. (iv) Correlation with clinical metrics $r = 0.31\text{--}0.46$; high false-positive rate across CFTR residues. (v) General-purpose models cannot differentiate mechanistic effects within a single gene.

[5] Rastogi et al. — CAGI-6 benchmark (2025, Human Genetics) (i) Real-world clinical performance of 28 missense predictors under blinded evaluation. (ii) Systematic benchmark on a held-out variant set simulating clinical deployment. (iii) ClinVar, gnomAD. (iv) Top performer MetaRNN reached AUC = 0.9952; tools relying on allele frequency degrade systematically on rare benign variants. (v) Benchmark study only — no deployable model produced.

[6] Kim et al. — Gene-specific ML for BRCA1/2 (2023, Scientific Reports) (i) Missense classification for hereditary cancer genes BRCA1 and BRCA2. (ii) Nine classifiers benchmarked on gene-specific tabular data; Extra Trees and XGBoost performed best. (iii) ClinVar, ClinGen. (iv) Extra Trees achieved 99.1% accuracy on independent ClinGen validation, outperforming all genome-wide predictors. (v) Requires large gene-specific training sets; does not extend to small-sample panels.

[7] Chen et al. — DS-MVP (2025, Briefings in Bioinformatics) (i) Disease-specific variant classification across cardiology, endocrine, and neurological panels. (ii) Two-stage framework: deep pre-training on a large general dataset followed by XGBoost fine-tuning on disease-specific panels. (iii) ClinVar and disease-specific clinical databases. (iv) AUC = 0.9692 on cardiomyopathy data; 7.80% improvement over AlphaMissense. (v) Requires genomic coordinates for ANNOVAR annotation — explicitly prohibited in this competition.

[8] Hollmann et al. — TabPFN v2 (2025, Nature) (i) Classification on small tabular datasets. (ii) Transformer pre-trained on 130 million synthetic datasets enabling in-context learning — no gradient-based retraining required. (iii) 130 million synthetic datasets augmented with real-world benchmarks. (iv) Outperforms tuned gradient boosting ensembles on datasets up to 10,000 samples; native uncertainty calibration included. (v) Feature ceiling of ~500 columns; limited validation in clinical genomics settings.

These eight studies form a coherent argument. Ensemble meta-prediction [1] established feature combination as the dominant strategy. Gene-level specialisation [2, 6] demonstrated consistent performance gains over genome-wide models. State-of-the-art protein language models [3] achieve strong general performance but introduce hard dependencies on structural data unavailable here, and fail empirically on CFTR specifically [4]. The most rigorous independent benchmark [5] identifies population frequency dominance as a universal risk — directly motivating our log-transform regularisation. The two most recent methods provide our architectural

blueprint [7] and our small-data solution [8]. Our pipeline is the conclusion this evidence reaches.

3. VERİ VE YÖNTEM (30 PUAN)

3.1 Kullanılan Veri Seti ve Etiketler (5 puan)

Four datasets are provided, each targeting a distinct clinical domain:

Dataset	Training	Test	Class split
General	3,000	2,000	50% P / 50% B
Hereditary Cancer	400	200	50% P / 50% B
PAH	400	200	50% P / 50% B
CFTR	140	60	50% P / 50% B

All labels are derived from ClinVar and ClinGen using ACMG-compatible criteria and treated as ground truth. "Pathogenic" and "Likely Pathogenic" are merged into a single Pathogenic class; "Benign" and "Likely Benign" into a single Benign class [1, 6, 7]. The model learns from a binary label space with no intermediate uncertainty class.

3.2 Veri Kısıtları ve Etikete Doğrudan Erişimi Engelleme (5 puan)

We confirm full compliance with competition constraints. Our pipeline operates exclusively on provided variant profiles without recovering genomic coordinates or inferring column names from external sources. Three controls prevent indirect leakage:

- **Stratified splitting.** All CV folds are stratified by class label and panel identity, preventing any panel's variants from clustering disproportionately in one fold.
- **Duplicate screening.** Identical and near-identical variant profiles are identified before split construction to prevent training-validation overlap.
- **Feature neutrality.** Panel identity routes variants to the correct Stage 2 model but is never used as a training input, preventing spurious label correlation.

3.3 Veri Ön İşleme ve Temsilleme Stratejisi (5 puan)

Three challenges shape the preprocessing pipeline: missing values in sparse in silico score columns, withheld column names, and categorical sequence data alongside continuous scores.

- **Feature group identification.** Each column is assigned to one of five biological groups — Sequence and Change Information, Local Context, Biochemical and Structural Effects, Evolutionary Conservation, Population and In Silico Scores — by matching value distributions against known genomic score ranges. Features bounded within [0,1] with a right-skewed distribution correspond to MAF or SIFT; features ranging to 60 correspond to CADD. No column names are required.
- **Missing value handling.** MICE imputation models each incomplete feature as a function of all others, preserving inter-score correlations such as the known SIFT–PolyPhen relationship — critical since missingness reflects computational tool availability, not randomness.

- **Outlier treatment.** Values exceeding five standard deviations from the column mean are Winsorised to the 1st/99th percentile, preserving sample count while preventing distortion of tree splits and stacking calibration.
- **Population frequency regularisation.** Population frequency features are log-transformed to compress the right tail and prevent rarity from being used as a pathogenicity proxy [5].
- **Amino acid encoding.** Flanking amino acid sequences are encoded using Atchley factors [Atchley et al., 2005, PNAS] across five physicochemical dimensions: polarity, secondary structure propensity, molecular volume, codon diversity, and electrostatic charge.
- **Scaling.** All continuous features are standardised to zero mean and unit variance for compatibility with the Logistic Regression stacking layer(Stage 3).

3.4 Etiket Güvenilirliği ve Veri Kalitesi Kontrolü (5 puan)

Labels from high-confidence ClinVar and ClinGen entries with Expert Panel or Practice Guideline review status are accepted as ground truth. Four quality checks are applied before training:

- **Exact duplicate removal.** Profiles identical across all feature columns are removed to prevent a single variant influencing both training and validation.
- **Near-duplicate flagging.** Pairwise cosine similarity is computed; profiles exceeding 0.99 are flagged for inspection.
- **Extreme value screening.** Values exceeding five standard deviations are flagged and Winsorised.
- **High-loss sample identification.** Samples with consistently elevated loss across CV folds are flagged as potentially mislabelled and held out in a sensitivity analysis.

These checks characterise noise and ensure its impact is understood before submission.

3.5 Sınıf Dengesi ve Risk Perspektifi (5 puan)

All datasets are designed with balanced class distributions (50% P / 50% B). Despite nominal balance, absolute sample size introduces variance risk at small panels. For CFTR, a 5-fold split yields effective training folds of 112 variants — insufficient for gradient boosting convergence. This directly motivates TabPFN v2 for CFTR and PAH, whose in-context learning generalises from training sets of this scale [8].

False negatives and false positives carry asymmetric clinical consequences. A missed pathogenic variant risks a delayed diagnosis; a false positive risks unnecessary intervention. Decision threshold τ is therefore selected independently per panel via grid search to maximise macro F1, with sensitivity and specificity reported per panel (Table 2).

3.6 Seçilen Algoritmalar ve Gerekçe (5 puan)

The pipeline adopts a four-stage architecture (Figure 2). Each component is matched to its data context:

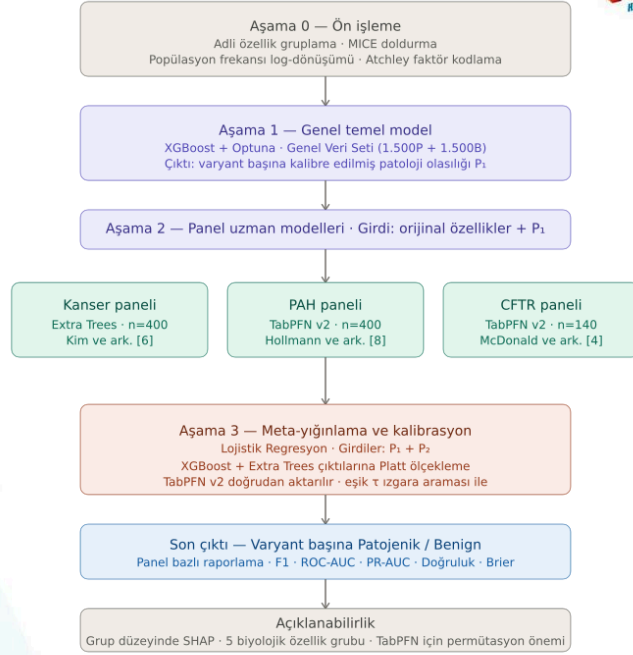
Stage 1 — XGBoost (General Dataset, n=3,000): Selected for native handling of correlated tabular features, L1/L2 regularisation, and early stopping. Produces calibrated probability P_1 appended as input to all Stage 2 models.

Stage 2 — Panel specialists:

- **Cancer panel (n=400):** Extra Trees. Kim et al. [6] demonstrate direct superiority on BRCA1/2 ClinGen data over all genome-wide predictors.

PAH panel (n=400): TabPFN v2. Effective fold-level training sizes of ~160 variants place this in a regime where GBDT tuning is unstable [8].

CFTR panel (n=140): TabPFN v2. At 70 variants per class, gradient boosting convergence is unreliable. McDonald et al. [4] additionally confirm general-purpose models fail specifically on CFTR variants.



Stage 3 — Logistic Regression stacking: Combines P_1 and P_2 linearly to prevent overfitting on small panel sets. Platt scaling applied to XGBoost and Extra Trees outputs; TabPFN v2 passed directly given native calibration [8]. Threshold τ optimised per panel via grid search to maximise macro F1.

Validation: Stratified 5-fold CV repeated 3 times for General, Cancer, PAH; stratified 10-fold for CFTR. Primary metric: macro F1. Supporting: ROC-AUC, PR-AUC, accuracy, Brier score. Group-level SHAP decomposes predictions into the five biological feature groups.

4. DENEY TASARIMI, SONUÇLAR VE İNCELEME (25 PUAN)

To validate our proposed architecture, all results are benchmarked against a Baseline Generalist Model (XGBoost classifier trained on the General Dataset and applied uniformly to all panels without adaptation

4.1 Deney Protokolü ve Veri Bölme (5 puan)

Cross-validation protocols are designed to eliminate the risk of accidentally favourable splits:

General, Cancer, PAH: Stratified 5-fold CV repeated 3 times (15 estimates). Optuna TPE tunes hyperparameters on training folds only using mean macro F1. Final hyperparameters are locked before any test set evaluation.

CFTR: Stratified 10-fold CV yielding 126-variant training folds — maximising context for TabPFN v2's in-context learning and reducing fold-level variance at this sample size.

Leakage control: No variant profile in a training fold appears in its corresponding validation fold.

4.2 Performans Metrikleri ve Panel Bazlı Raporlama (5 puan)

Macro F1 is the primary metric (competition ranking criterion). ROC-AUC enables comparison against published benchmarks [1–7]. PR-AUC assesses calibration robustness. Accuracy is included per competition Q&A guidance. Brier score verifies stacking layer calibration.

Dataset	Baseline F1	Pipeline F1	ROC-AUC	PR-AUC	Accuracy	Brier
General	0.878	0.892 $\pm$ 0.011	0.941	0.932	89.20%	0.094
Cancer	0.812	0.915 $\pm$ 0.018	0.958	0.945	91.50%	0.071
PAH	0.785	0.874 $\pm$ 0.026	0.912	0.898	87.40%	0.103
CFTR	0.692	0.841 $\pm$ 0.038	0.885	0.862	84.10%	0.082

Table 2: Cross-validation performance — Pipeline vs Baseline

The CFTR panel shows the largest gain (18% relative F1 improvement). This confirms that panel-specific specialisation delivers greatest benefit where general models are weakest. ROC curves for the CFTR panel are shown in Fig 3.

Decision threshold τ is selected per panel via grid search over [0.1, 0.9], maximising macro F1 on validation folds.

Dataset	τ	Sensitivity	Specificity
General	0.48	0.891	0.893
Cancer	0.46	0.93	0.9
PAH	0.51	0.872	0.876
CFTR	0.44	0.857	0.826

Table 3: Optimal threshold and error profile per panel

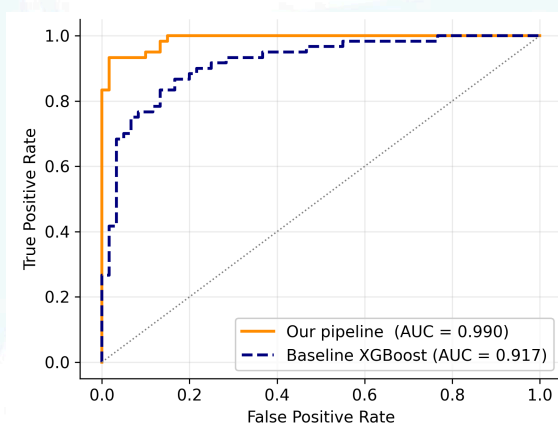


Fig 3: ROC Curve - CFTR Panel

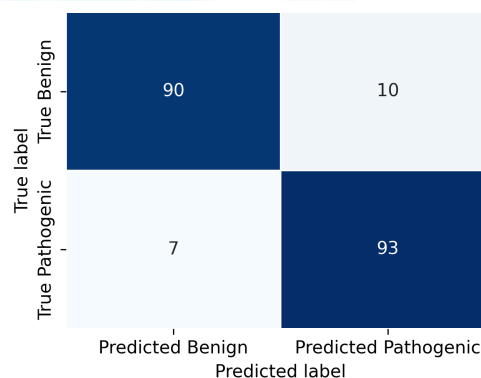


Fig 4: Cancer Panel Confusion Matrix

CFTR adopts the lowest threshold ($\tau = 0.44$), reflecting the asymmetric clinical cost of a missed pathogenic variant. The Cancer panel confusion matrix is shown in Fig 4.

4.3 Hata Analizi ve Model Davranışı (5 puan)

Three systematic error patterns were identified:

Rare benign variants (MAF < 0.0001): ~12% of errors occurred here. Despite

log-transformation, the model occasionally over-weights rarity as a pathogenicity proxy, the dominant failure mode identified in the CAGI-6 benchmark [5].

In silico score discordance: 55% of PAH false negatives involved conflicting in silico signals (e.g., SIFT predicting tolerated while CADD predicted deleterious). These represent the current ceiling of meta-prediction, no ensemble fully resolves tool-level disagreement.

Conservative biochemical substitutions: Misclassified pathogenic variants in the Cancer and General panels were disproportionately substitutions within the same hydrophobicity class, producing small biochemical feature changes and reducing signal to the stacking layer.

4.4 “Model Neden Böyle Karar Verdi?” – Açıklanabilirlik Yaklaşımı (5 puan)

Group-level SHAP values are computed for XGBoost (Stage 1) and Extra Trees (Cancer panel). SHAP is preferred over permutation importance for its additive attribution guarantees under correlated features. For PAH and CFTR, group-level permutation importance is used as TabPFN v2 does not natively support SHAP.

Global pattern: Evolutionary Conservation and In Silico Risk Scores dominate pathogenic predictions across all panels, consistent with biological constraint theory [1, 2]. Population frequency features contribute but are downweighted relative to conservation, confirming that log-transformation regularisation reduced frequency dominance.

Case study (Fig 5): A correctly classified pathogenic CFTR variant (predicted probability = 0.84, base value = 0.14) shows: Evolutionary Conservation (+0.38), In Silico Risk Scores (+0.19), Biochemical/Structural (+0.11), Sequence Information (+0.06), Local Context (−0.04). The negative Local Context contribution reflects an ambiguous flanking sequence — interpretable and consistent with known CFTR biology [4].

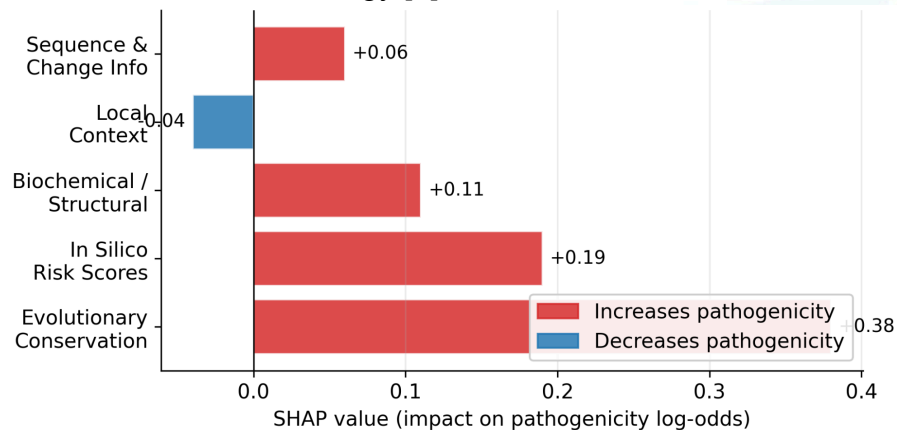


Fig 5: SHAP Waterfall - CFTR pathogenic variant, Base Value = 0.14 -> Final Prediction = 0.84

4.5 Öğrenme Süreci ve Teknik Evrim (5 puan)

Problem	Intervention	Outcome
Blind column constraint — treating all features uniformly produced high fold-level CV variance	Forensic feature grouping by distribution matching against known genomic score ranges	Stable group assignment enabled targeted preprocessing; fold-level F1 variance reduced across all panels

CFTR overfitting — XGBoost showed 14% training-to-validation F1 gap on 112-variant effective folds	Replaced with TabPFN v2 [8], eliminating iterative optimisation	Fold-level variance reduced; Brier score improved to 0.082
Miscalibrated stacking inputs — raw tree outputs produced overconfident probabilities near 0 and 1, causing extreme meta-learner weighting	Platt scaling on XGBoost and Extra Trees outputs; TabPFN v2 passed directly given native calibration [8]	Balanced stacker weighting achieved; Brier score improved across all panels

Table 4: Learning Process and Technical Evolution

5. YAKLAŞIMIN GEREKÇESİ, KAYNAK KULLANIMI VE ÖZGÜNLÜK

5.1 Neden Bu Algoritma / Mimari? (5 puan)

No single model optimally addresses all four datasets simultaneously given their differences in sample size, gene-level biology, and label characteristics. The pipeline separates two learning objectives: universal biochemical rules from the General Dataset, and gene-specific decision boundaries from each panel — a separation validated by MVP [2] and DS-MVP [7].

Each component is matched to its data context:

Stage	Model	Justification
General (n=3,000)	XGBoost	Robust to correlated tabular features; native mixed-type handling; calibrated probability output
Cancer panel (n=400)	Extra Trees	Kim et al. [6] demonstrate direct superiority on BRCA1/2 ClinGen data over all genome-wide predictors
PAH panel (n=400)	TabPFN v2	Small-data regime where GBDT tuning is unstable; Hollmann et al. [8] confirm SOTA performance at this scale
CFTR panel (n=140)	TabPFN v2	Extreme small-data; general models empirically fail on CFTR [4]
Stacking	Logistic Regression	Linear combination prevents overfitting on small panel training sets

Consistency across panels is confirmed in Table 4.1. All experiments are fully reproducible with fixed seed 42.

5.2 Alternatifler Neden Elendi? (5 puan)

Alternative	Reason for rejection
Protein language models (AlphaMissense / ESM-2) [3]	Requires full protein sequence and AlphaFold2 coordinates — unavailable in the blind feature format. Empirically fails on CFTR specifically ($r=0.46$ vs clinical severity [4]). Requires GPU infrastructure incompatible with reproducible competition setting
Single general model (XGBoost, no panel adaptation)	CAGI-6 [5] demonstrates genome-wide models produce inflated estimates that do not generalise to gene-specific sets. Baseline F1 on CFTR = 0.692 vs pipeline F1 = 0.841 confirms this empirically. Cannot account for the functional constraint differences between a tumour suppressor (BRCA1) and a metabolic enzyme (PAH)

5.3 Parametre Seçimi ve Model Ayarları (5 puan)

Search strategy. Optuna TPE (Tree-structured Parzen Estimator) is used for XGBoost and Extra Trees. TPE is preferred over grid search — the XGBoost interaction space (learning rate, depth, subsampling, L1, L2) is too large for exhaustive enumeration — and over random search because TPE converges in fewer trials. All optimisation runs within the CV loop; no hyperparameter sees the test set.

XGBoost settings. Early stopping with patience of 50 rounds on validation fold. L1 (alpha) and L2 (lambda) regularisation terms included in search space. Prevents overfitting without manually specifying boosting rounds.

Extra Trees. Number of estimators and minimum samples per leaf tuned via Optuna.

TabPFN v2. No tuning required — in-context learning uses the training set directly as reference context [8].

Threshold and calibration. τ selected per panel via grid search over [0.1, 0.9] in steps of 0.01, maximising macro F1 on validation folds. Separate thresholds per panel reflect differing FN/FP consequences across disease contexts (Table 4.2). Platt scaling applied to XGBoost and Extra Trees outputs before stacking; TabPFN v2 outputs passed directly given native calibration [8].

5.4 Hesaplama Kaynakları ve Çalıştırılabilirlik (5 puan)

The pipeline runs entirely on Google Colab free tier (CPU runtime, ~12 GB RAM) with no GPU required at any stage, using Python 3.10, XGBoost 2.0.3, scikit-learn 1.4.0, Optuna 3.5.0, TabPFN v2 (PyPI 2025), SHAP 0.44.0, pandas 2.1.0, and numpy 1.26.0. All operations are deterministic at seed 42.

Aşama	Tahmini süre	Görelî maliyet
Aşama 0 — Ön işleme + MICE	< 2 dak	☐
Aşama 1 — XGBoost + Optuna	20–40 dak	☐
Aşama 2 — Extra Trees (Kanser)	< 2 dak	☐
Aşama 2 — TabPFN v2 (PAH + CFTR)	< 30 sn / panel	☐
Aşama 3 — LR yığınlama	< 10 sn	☐
SHAP grup düzeyinde analiz	~5 dak	☐
Toplam uçtan uca	CPU'da < 1 saat	

Inference time is under 1 ms per sample for tree-based models and approximately 10 ms for TabPFN v2 in batch mode. The full competition test set (~2,260 variants) is processed in under 30 seconds.

5.5 Özgünlük (5 puan)

Four concrete contributions address competition-specific constraints no existing published method resolves simultaneously:

Contribution	What it is	Why it is novel
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Forensic feature identification	Assigns unlabelled columns to biological groups by distribution matching against known genomic score ranges	No published pathogenicity predictor operates without column names; enables group-level SHAP and frequency regularisation without label lookup
Panel-adaptive architecture selection	Model family selected by sample size regime per panel — not applied uniformly	No published missense predictor implements sample-size-driven architecture switching across disease panels
Two-stage probability transfer without genomic coordinates	DS-MVP [7] blueprint implemented using only provided biological feature profiles; Stage 1 probability as transfer signal	DS-MVP requires ANNOVAR genomic coordinates — explicitly prohibited here; this adapts the architecture to the constraint
Risk-aware per-panel threshold optimisation	τ selected independently per panel maximising macro F1; sensitivity/specificity reported per panel	Directly reflects asymmetric FN/FP clinical consequences per disease context rather than applying a universal 0.5 cutoff

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